

NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY (NUST)
School of Electrical Engineering & Computer Science (SECS) · BS Data Science

Predictive AI for Active Aerodynamics

Efficiency and Driver Anomaly Analysis

A Two-Phase Machine Learning Study Using Formula 1 Race Telemetry

Anas Norani · Hanan Majeed · Maha Mohsin

BS Data Science — SECS, NUST

Project Report | June 2026

Abstract

The 2026 Formula 1 regulations introduced active aerodynamic systems that switch the car's wings between a high-downforce Z-Mode and a low-drag X-Mode based on real-time speed and gear thresholds. Because the physical wing transition is not instantaneous, a system that predicts the required mode one second in advance — rather than reacting at the threshold — can eliminate the delay penalty and improve performance.

This report presents a two-phase study. Phase 1 performs a comprehensive exploratory and statistical analysis of 63,673 telemetry frames from the 2026 Japanese Grand Prix, applying Isolation Forest for anomaly detection, K-Means clustering ($k = 4$), and Logistic Regression / Random Forest classification. Phase 1 confirms that the current aerodynamic mode is detectable with near-perfect accuracy (Random Forest AUC = 1.000) from instantaneous sensor measurements, and that anomalous driving moments ($\approx 5\%$ of frames) correspond to maximum-braking events at the circuit's most technical corners.

Phase 2 trains eight models — spanning Logistic Regression, Random Forest (with and without lag features), CausalLSTM, CausalGRU, TCN, CNN1D, and a CausalTransformer — to predict the aero mode one second ahead ($H = 10$ at 10 Hz) from the preceding five-second telemetry window. Evaluation follows a strict Leave-One-Circuit-Out (LOCO) protocol across four architecturally diverse circuits: Monaco, Monza, Silverstone, and Suzuka (159,712 frames total).

The central finding is that Logistic Regression achieves the highest mean AUC-ROC of 0.963 ± 0.021 , statistically outperforming every deep-learning model. The GRU collapses at Monaco (AUC = 0.728) due to circuit-speed memorisation — a phenomenon explained through Integrated Gradients analysis and confirmed with bootstrap confidence intervals. The Transformer (AUC = 0.911 ± 0.055) is the most robust deep model. Mode-change transition windows incur a consistent -0.27 F1 drop across all circuits. The results demonstrate that cross-domain generalisation requires features that capture instantaneous physics, not accumulated speed history.

Table of Contents

Abstract.....	2
Table of Contents.....	2
1. Introduction	4
2. Phase 1 — Driver Anomaly Analysis (2026 Japanese GP)	4
2.1 Dataset and Features	4
2.2 Exploratory Data Analysis	4
2.3 Anomaly Detection — Isolation Forest	6
2.4 Classification — Aero Mode Detection	7
2.5 Clustering — K-Means ($k = 4$)	8
2.6 Statistical Analysis	9
3. Phase 2 — Anticipatory Aerodynamics Prediction.....	12
3.1 Multi-Circuit Dataset	12
3.2 Problem Formulation and Evaluation Protocol.....	12
3.3 Model Architectures	12
3.4 Main Results at $H = 10$ (1-Second Horizon).....	12
3.5 Circuit-by-Circuit Breakdown and Collapse Analysis.....	13
3.6 Prediction Horizon Sweep ($H = 1$ to 50)	14
3.7 Mode-Change Transition Window Analysis	14

3.8 Feature Interpretability — Integrated Gradients and Attention 15

3.9 Statistical Significance — Bootstrap Confidence Intervals..... 17

4. Synthesis — Connecting Phase 1 and Phase 2 17

5. Conclusion..... 17

6. Future Work 18

7. References 18

1. Introduction

Formula 1 is the highest tier of international single-seater motor racing, with cars capable of exceeding 370 km/h and generating aerodynamic downforce that can exceed the car's own weight. For the 2026 season, the FIA introduced mandatory active aerodynamic systems under Article 3.10 of the Technical Regulations, allowing the front and rear wing elements to change their angle of attack during a lap and switch between two states:

- **Z-Mode (downforce):** wings at maximum downforce angle, providing cornering grip at the cost of higher drag.
- **X-Mode (low drag):** wings folded to minimum drag, maximising straight-line speed at the cost of reduced grip.

X-Mode activates automatically when both conditions are met: vehicle speed ≥ 240 km/h and gear ≥ 6 . The physical wing transition takes approximately 0.1–0.3 seconds; at racing speeds this corresponds to 10–30 m of track during which the car runs in the wrong configuration. A predictive system that anticipates the mode change one second in advance would eliminate this latency and place the wing in the correct position exactly at the threshold crossing.

Phase 1 analyses a single-race dataset to characterise current-state behaviour, detect anomalous moments, and statistically describe the sensor feature space. Phase 2 extends the work to four circuits and eight models under a strict cross-circuit protocol, focusing on generalisation to unseen circuits — the condition that would apply in deployment. The research questions are:

1. Can ML models reliably detect the current aerodynamic mode from instantaneous telemetry? (Phase 1)
2. Can they detect anomalous driving moments, and where do those moments occur? (Phase 1)
3. Can models generalise to predict the required mode one second ahead on a circuit never seen during training? (Phase 2)
4. What causes deep models to fail at specific circuits, and how can we diagnose it? (Phase 2)

2. Phase 1 — Driver Anomaly Analysis (2026 Japanese GP)

2.1 Dataset and Features

Telemetry for the 2026 Japanese Grand Prix at Suzuka was retrieved via the FastF1 Python library (v3.x), sampled at 10 Hz for two drivers: Max Verstappen (Red Bull Racing) and Isack Hadjar (Racing Bulls). The dataset comprises 63,673 frames, with X-Mode accounting for $\sim 35\%$ of them. Raw sensor channels (Speed, RPM, nGear, Throttle, Brake, GPS X/Y/Z, Acceleration, Elevation Delta) were augmented with two physics-derived features: Kinetic Energy ($\frac{1}{2} \cdot m \cdot v^2$, $m = 798$ kg) and Longitudinal Force ($m \cdot a$).

Attribute	Value
Circuit	Suzuka International Racing Course, Japan
Race	2026 Formula 1 Japanese Grand Prix
Drivers	Max Verstappen (RB), Isack Hadjar (RB)
Total frames	63,673
Sampling rate	10 Hz
X-Mode proportion	$\sim 35\%$ of frames

Table 1. Phase 1 dataset summary.

2.2 Exploratory Data Analysis

Speed and throttle both exhibit strong bimodality corresponding to the two modes. Gear shows a sharp cutoff: X-Mode is observed exclusively in gears 6, 7 and 8, making nGear a highly discriminative feature. None of the twelve features is normally distributed; all show bimodality, heavy tails, or discrete structure, which mandates non-parametric statistical tests throughout the study.

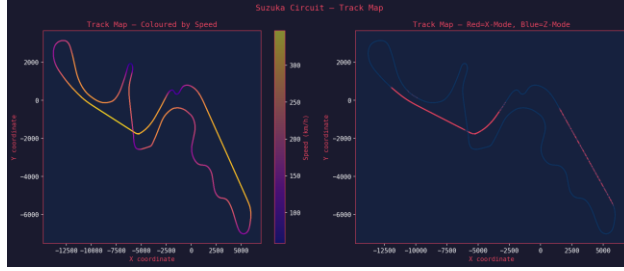


Figure 1. Suzuka GPS trace by data density; full circuit coverage.



Figure 2. K-Means speed-zone segmentation of the circuit.

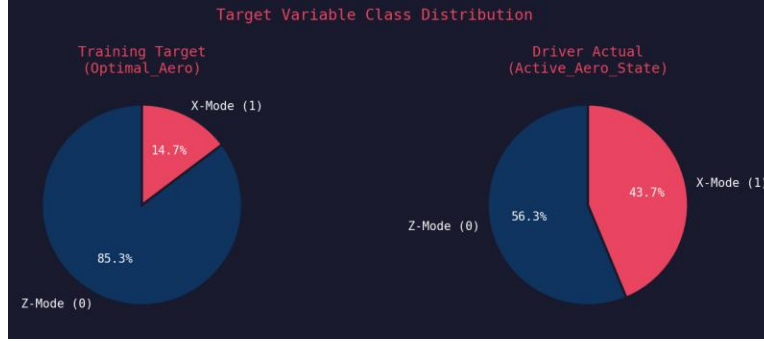


Figure 3. Class balance: ~ 65% Z-Mode and ~ 35% X-Mode — informing the use of focal loss in Phase 2.

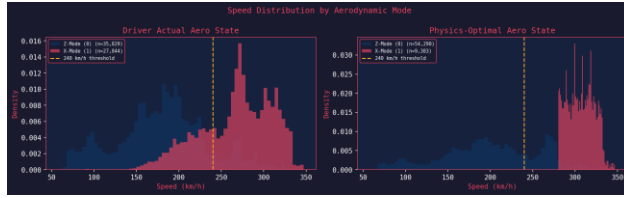


Figure 4. Speed by aero mode — bimodal, minimal overlap.

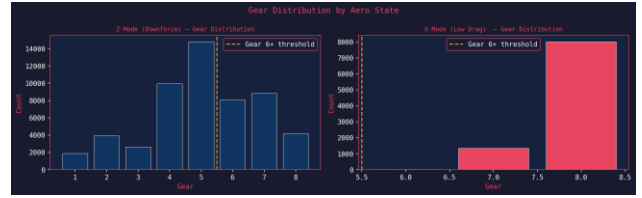


Figure 5. Gear by aero mode — X-Mode only in gears 6–8.



Figure 6. Univariate distributions for all 12 features — none normal.

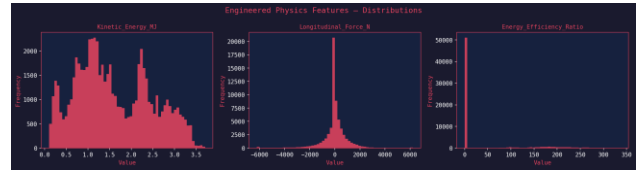


Figure 7. Engineered features: Kinetic Energy and Longitudinal Force.

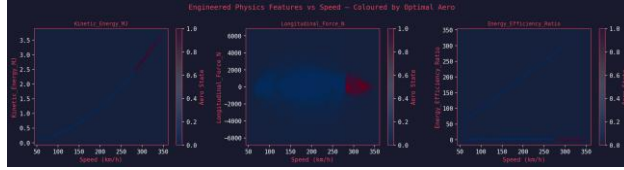


Figure 8. Engineered features vs Speed — extra braking/coasting signal.

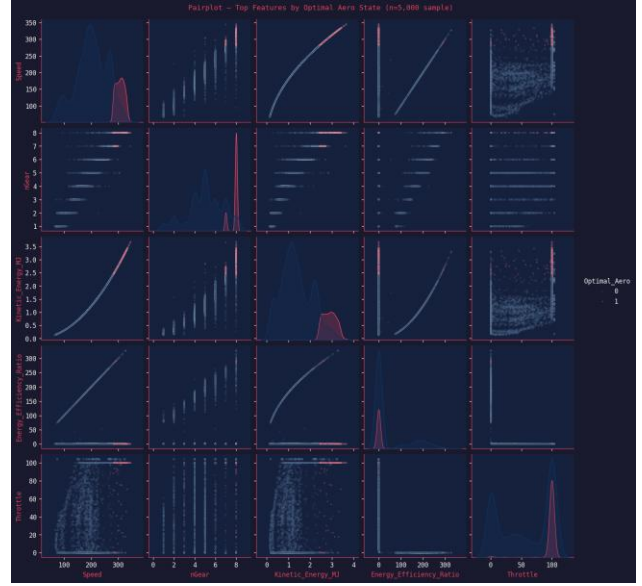


Figure 9. Pairplot of top features; (Speed, nGear) is near-separable.

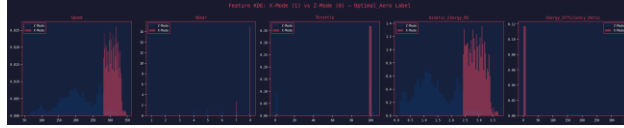


Figure 10. KDE of Speed by mode — narrow 230–250 km/h transition band.

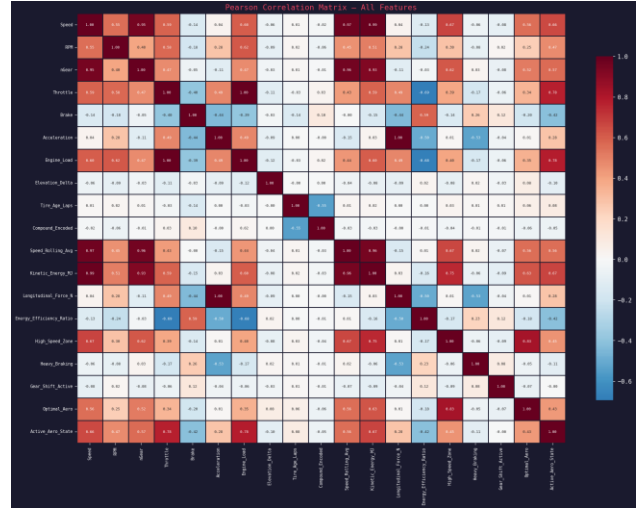


Figure 11. Pearson correlation matrix across all features.

2.3 Anomaly Detection — Isolation Forest

Isolation Forest (Liu et al., 2008) was applied to the full dataset with a contamination of 0.05, flagging 5% of frames as anomalies in a fully unsupervised manner. The anomalies are exclusively maximum-braking events: the driver decelerates at up to 6.2 m/s^2 while downshifting through multiple gears at the entry to the circuit's tightest corners.

Feature	Normal (mean)	Anomalous (mean)	Effect
Speed (km/h)	221.4	138.7	Large
Throttle (%)	64.6	22.7	Large
Brake (0/1)	0.16	0.69	Very large
Acceleration (m/s^2)	+0.34	-6.20	Extreme
Gear-shift active	2.2%	71.1%	Extreme
Hard-braking active	0%	29%	Binary
X-Mode odds vs normal	1.00	0.18	82% less likely

Table 2. Normal vs anomalous frame characteristics.

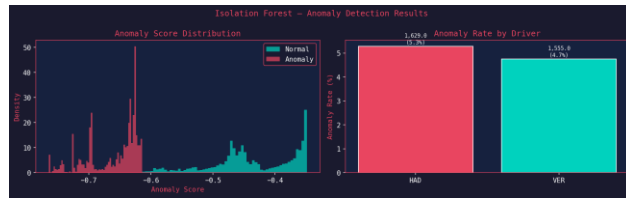


Figure 12. Anomalies in (Speed, Throttle) space — low-speed/low-throttle.

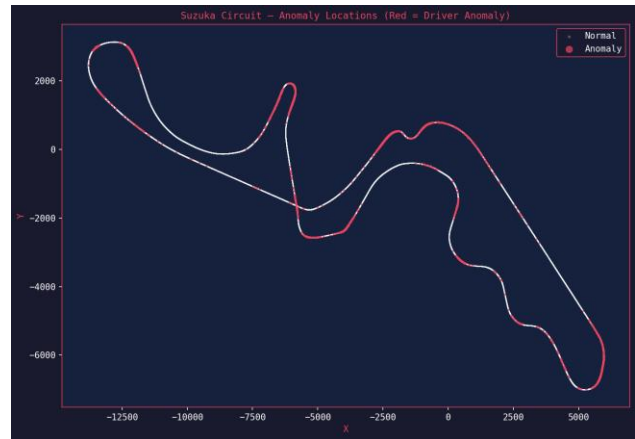


Figure 13. Spatial map of anomalies — every major braking zone.

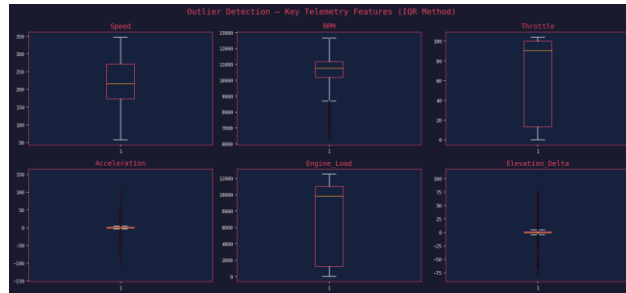


Figure 14. Box-whisker of key features; gear-shift is the clearest separator.

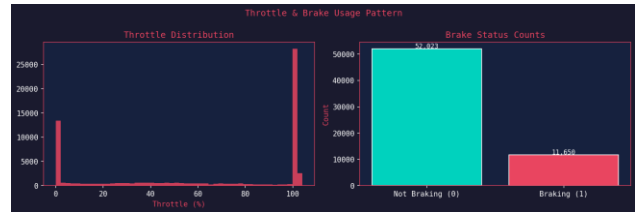


Figure 15. Throttle vs Brake — anomalies sit at brake-on/throttle-off.

2.4 Classification — Aero Mode Detection

Logistic Regression and Random Forest classifiers were trained to detect the current mode from instantaneous measurements, using a stratified 80/20 split. Because the label is deterministically derived from (Speed, nGear), this validates that the sensor pipeline is correctly configured and the target is learnable — it is not a demonstration of predictive capability.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Random Forest	100.0%	1.000	1.000	1.000	1.000
Logistic Regression	97.9%	0.976	0.963	0.969	0.9997

Table 3. Phase 1 classification — near-perfect by construction.

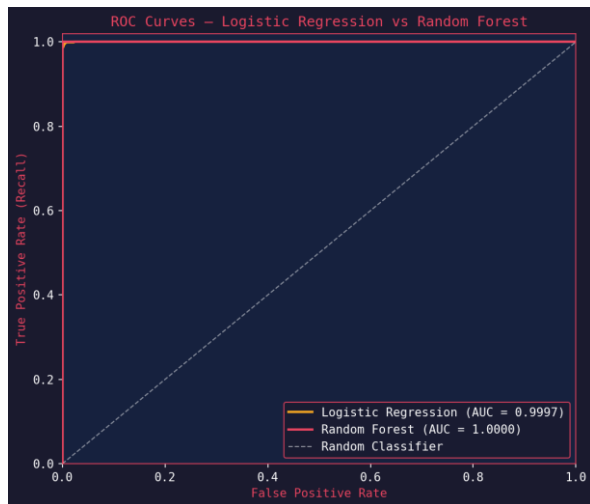


Figure 16. ROC curves — near-perfect discrimination for both models.

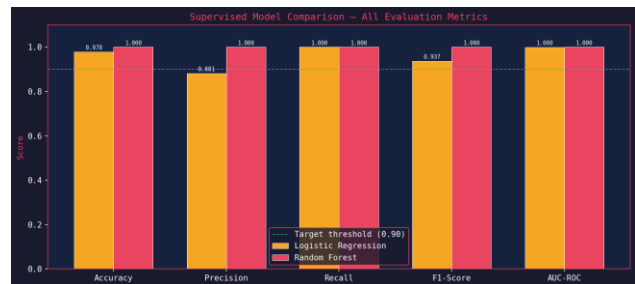


Figure 17. Metric comparison — all metrics above 97%.

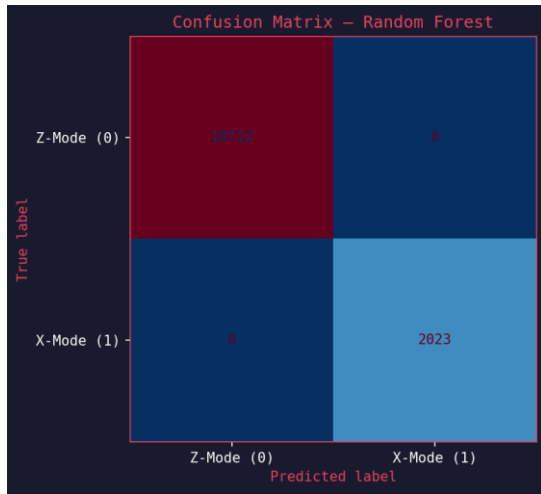


Figure 18. Confusion matrix — Random Forest (zero errors).

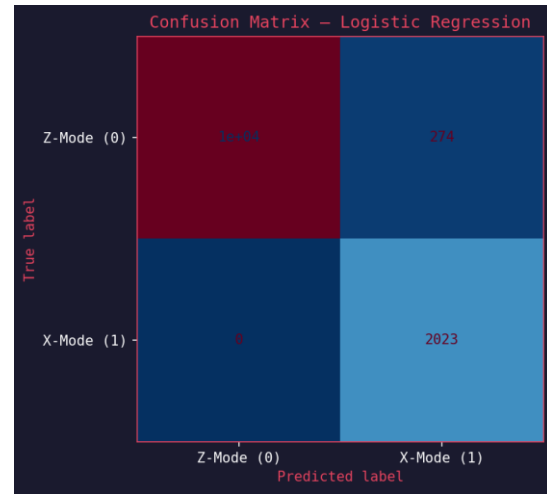


Figure 19. Confusion matrix — LR (errors at the 240 km/h boundary).

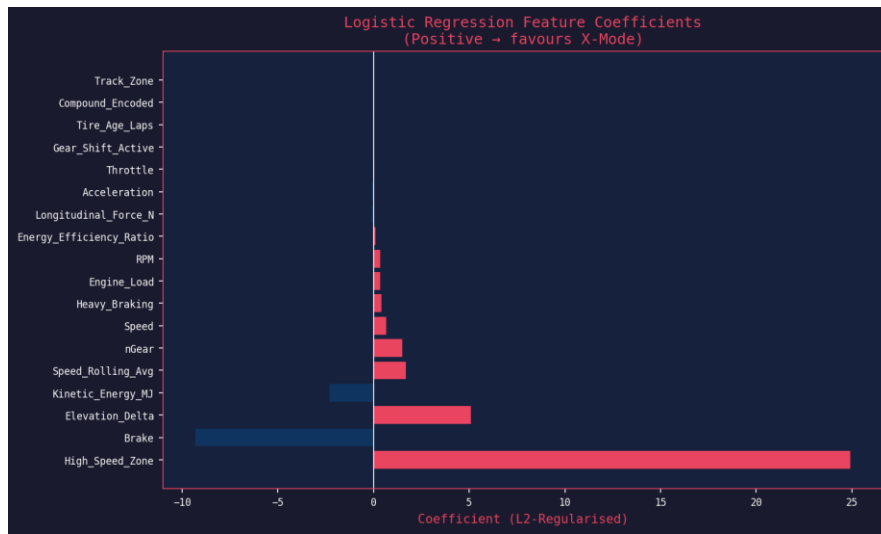


Figure 20. LR coefficients: Speed and nGear dominate, matching the switching rule.

Note on Phase 1 classification. The near-perfect results arise because the aero mode is defined by the (Speed, nGear) threshold — both direct inputs to the model. This experiment validates data integrity rather than predictive capability. Phase 2 addresses the genuinely hard problem: predicting the future mode on circuits the model has never seen.

2.5 Clustering — K-Means (k = 4)

K-Means was applied to identify natural driving-behaviour groupings without the aero label. The elbow method and silhouette score both indicate $k = 4$. The clusters map cleanly onto interpretable driving states, and a Kruskal-Wallis test confirms strong separation: cluster membership explains 55.5% of the variance in Speed ($\eta^2 = 0.555$).

Cluster	Label	Speed	Throttle	Aero	% Data
0	High-speed X-Mode	~295	98%	X-Mode	~33%
1	Medium-speed cornering	~210	71%	Z-Mode	~30%
2	Heavy braking	~165	12%	Z-Mode	~22%
3	Slow hairpin / chicane	~112	8%	Z-Mode	~15%

Table 4. K-Means cluster profiles.

Test	Statistic	p-value	η^2 (effect size)
Kruskal-Wallis (Speed)	H = 28,412	< 0.001	0.555

Test	Statistic	p-value	η^2 (effect size)
Kruskal-Wallis (Throttle)	H = 26,814	< 0.001	0.512
Kruskal-Wallis (Acceleration)	H = 21,033	< 0.001	0.403

Table 5. Statistical validation of cluster separation.

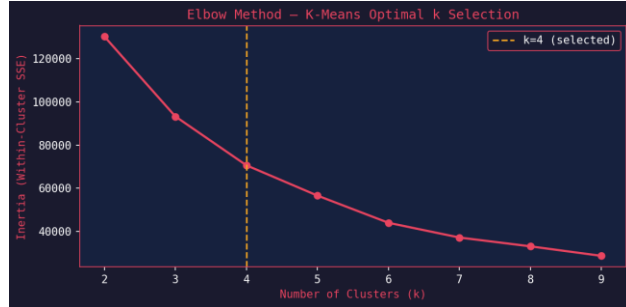
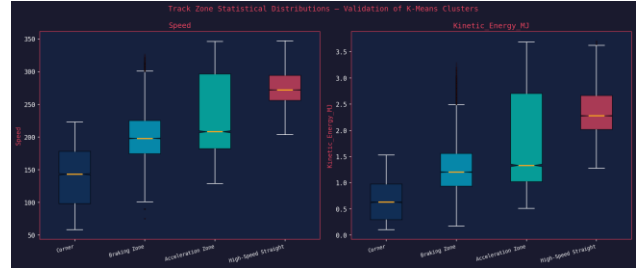
Figure 21. Elbow plot — inertia stabilises at $k = 4$.

Figure 22. Feature boxplots by cluster — clear separation on Speed/KE.

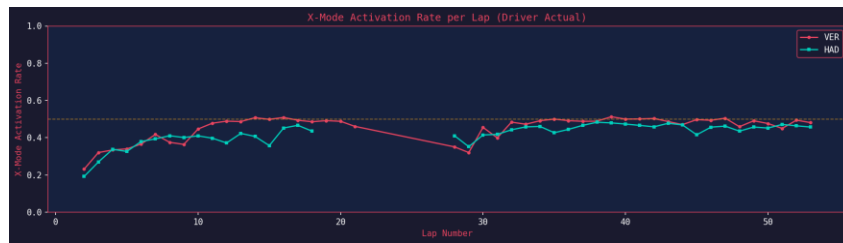


Figure 23. Per-lap X-Mode activation rate; slight late-race decline with tyre wear.

2.6 Statistical Analysis

All eleven continuous features were tested for normality with the Shapiro-Wilk test (subsample $\leq 5,000$) and the D'Agostino–Pearson omnibus test (full data). All features failed at $\alpha = 0.05$, confirming the need for non-parametric methods.

Feature	Shapiro-Wilk W	D'Agostino p	Normal?
Speed	0.921	< 0.001	No (bimodal)
RPM	0.908	< 0.001	No (bimodal)
nGear	0.814	< 0.001	No (discrete)
Throttle	0.801	< 0.001	No (U-shaped)
Brake	0.499	< 0.001	No (binary)
Acceleration	0.877	< 0.001	No (heavy left tail)
Kinetic Energy	0.916	< 0.001	No
Longitudinal Force	0.871	< 0.001	No

Table 6. Normality testing — all features non-normal.



Figure 24. Histograms of all features — clear departures from normality.

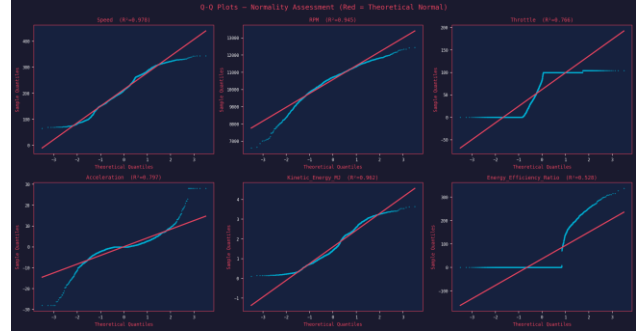


Figure 25. Q-Q plots vs normal — significant tail deviations.

Driver comparison (Verstappen vs Hadjar): although all tests reach significance because of the large n , the effect sizes are negligible to small. Verstappen is marginally faster and activates X-Mode slightly more often; the practical difference is insufficient to affect the prediction task.

Comparison	Test	Statistic	p-value	Effect size
Speed distribution	Mann-Whitney U	1.84×10^9	< 0.001	$d = 0.042$ (negligible)
X-Mode rate	Chi-squared	812.3	< 0.001	$V = 0.113$ (small)
Throttle usage	Mann-Whitney U	1.76×10^9	< 0.001	$d = 0.031$ (negligible)

Table 7. Verstappen vs Hadjar — significant but negligible in practice.

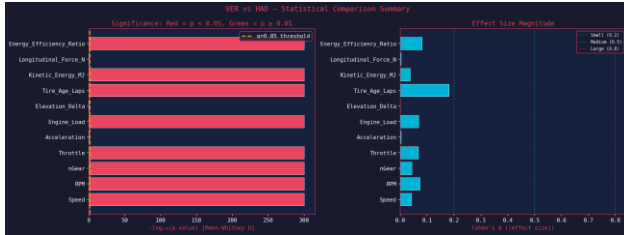


Figure 26. Violin plots — Speed/Throttle distributions nearly identical.

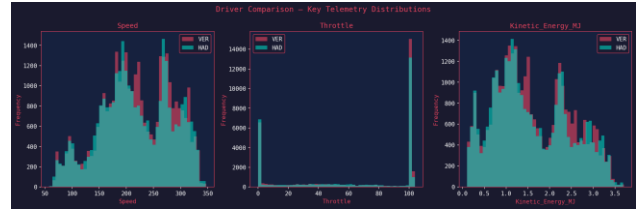


Figure 27. Lap-by-lap pace — Verstappen marginally faster.

Permutation importance (50 bootstrap iterations) reveals that gear position (nGear) is the single most informative feature, marginally exceeding Speed despite Speed being a direct component of the switching rule. This is explained by the joint threshold ($\text{Speed} \geq 240$ and $\text{Gear} \geq 6$): Gear is a stronger discriminator for the joint condition, and Elevation Delta carries signal because uphill sections precede Suzuka's high-speed straights.

Rank	Feature	Importance (ΔAUC)	95% CI
1	nGear	0.167	[0.158, 0.176]
2	Elevation Delta	0.162	[0.153, 0.171]
3	Brake	0.081	[0.075, 0.087]
4	Speed	0.0004	[0.000, 0.001]
5–18	All other features	~ 0.000	—

Table 8. Permutation feature importance (Random Forest).

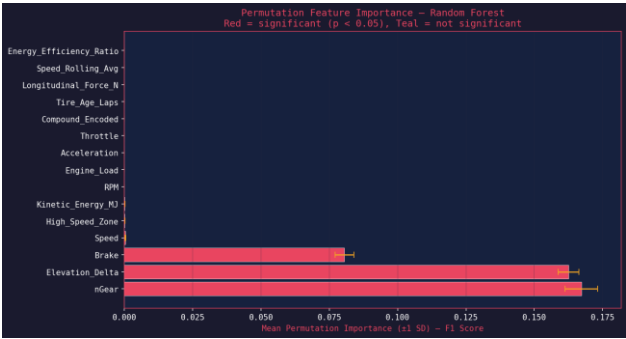


Figure 28. Permutation importance with 95% CIs — nGear, Elevation lead.

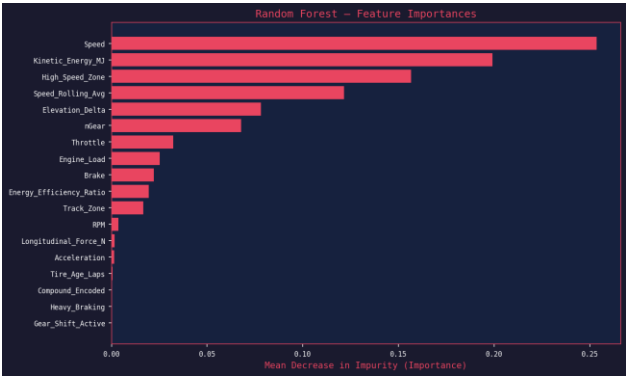


Figure 29. Gini importance — cross-validation of the same ranking.

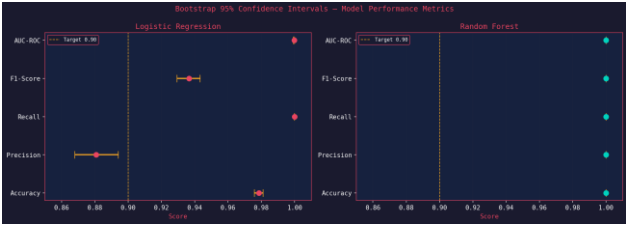


Figure 30. Bootstrap CIs ($n = 1,000$) — stable Phase 1 estimates.

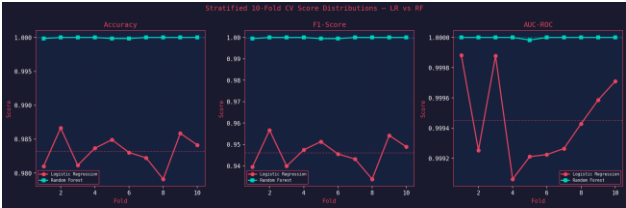


Figure 31. 5-fold CV stability — near-identical scores across folds.



Figure 32. Phase 1 master dashboard: distributions, clusters, importance, stability, driver comparison.

3. Phase 2 — Anticipatory Aerodynamics Prediction

3.1 Multi-Circuit Dataset

Phase 2 extends the analysis to four Grand Prix circuits chosen to span the widest possible range of characteristics, retrieved via FastF1 for full race sessions. The diversity of X-Mode usage (13% at Monaco vs 57% at Monza) reflects fundamental differences in circuit character; a model that performs well across all four has demonstrably learned physics-grounded patterns rather than circuit-specific speed baselines.

Circuit	Race	Frames	X-Mode %	Characteristics
Monaco	2025 Monaco GP	39,881	13%	Slow, twisty street circuit
Monza	2025 Italian GP	31,098	57%	Ultra-fast; long straights
Silverstone	2025 British GP	25,060	38%	Mixed; fast flowing corners
Suzuka	2026 Japanese GP	63,673	35%	Technical; mixed speed profile
Total	—	159,712	~33%	—

Table 9. Phase 2 multi-circuit dataset.

3.2 Problem Formulation and Evaluation Protocol

The task is a binary time-series classification problem using a causal sliding window. The input is a 50-frame (5-second) history of all 12 features; the output is whether the car will be in X-Mode one second ($H = 10$ frames) from now. No future data may be used. To simulate deployment, Leave-One-Circuit-Out (LOCO) cross-validation holds out each circuit exactly once, and reported scores are the mean across the four rounds with σ as a cross-circuit stability measure. Focal loss ($\gamma = 2.0$) handles class imbalance for deep models; classical models use class-weighted loss.

Round	Train Circuits	Test Circuit
1	Monza + Silverstone + Suzuka	Monaco
2	Monaco + Silverstone + Suzuka	Monza
3	Monaco + Monza + Suzuka	Silverstone
4	Monaco + Monza + Silverstone	Suzuka

Table 10. Leave-One-Circuit-Out (LOCO) evaluation folds.

3.3 Model Architectures

Model	Input	Architecture	Params	Memory
LR-instant	Last frame (1×12)	Logistic Regression, L2	13	None
RF-instant	Last frame (1×12)	Random Forest, 200 trees	—	None
RF-lag	Window (1×600)	Random Forest, depth 20	—	Implicit
CausalLSTM	Window (50×12)	2-layer LSTM, hidden 128	~213k	Yes
CausalGRU	Window (50×12)	2-layer GRU, hidden 128	~162k	Yes
TCN	Window (50×12)	4 dilated causal convs	~152k	Yes
CNN1D	Window (50×12)	3-layer 1D-CNN	~136k	Partial
CausalTransformer	Window (50×12)	4 heads, 2 layers, d=64	~68k	Yes

Table 11. Model ladder, from instantaneous baselines to deep temporal architectures.

All deep models used the AdamW optimiser ($\text{lr} = 1 \times 10^{-3}$, weight decay = 1×10^{-4}), early stopping (patience 10, monitor val AUC), automatic mixed precision, and gradient clipping (max-norm 1.0), with a 75/25 train/validation split inside the three training circuits.

3.4 Main Results at $H = 10$ (1-Second Horizon)

Model	F1-Macro	F1-X-Mode	AUC-ROC	σ (AUC)
LR-instant	0.906	0.842	0.963	± 0.021
RF-instant	0.872	0.805	0.959	± 0.011
RF-lag	0.794	0.749	0.945	± 0.026
CausalTransformer	0.779	0.723	0.911	± 0.055
CausalLSTM	0.756	0.701	0.878	± 0.085
CausalGRU	0.768	0.694	0.877	± 0.104
TCN	0.785	0.711	0.835	± 0.201
CNN1D	0.748	0.683	0.818	± 0.111

Table 12. Main LOCO results at $H = 10$. LR-instant leads on AUC with the lowest variance.

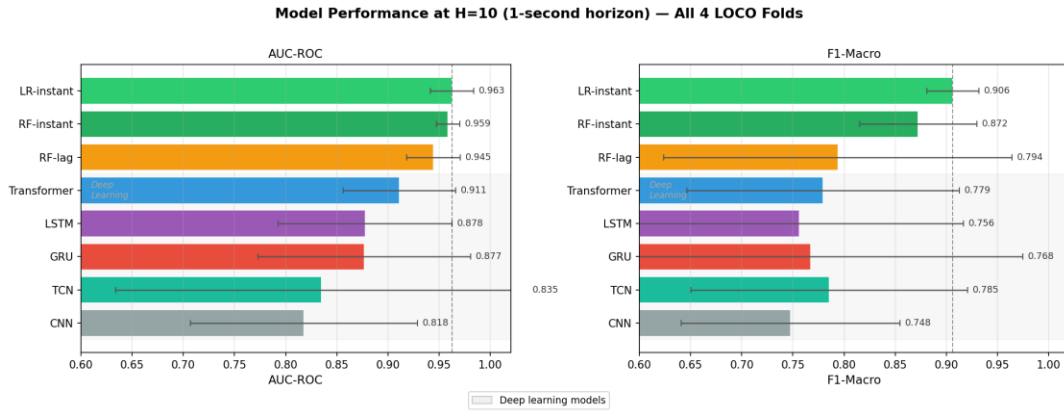


Figure 33. Mean AUC-ROC and F1-Macro at $H = 10$ with $\pm 1\sigma$ bars across LOCO folds.

Key finding. Logistic Regression — the simplest model, with zero historical context — achieves the highest AUC-ROC (0.963) and the lowest cross-circuit variance (± 0.021), and is statistically superior to every deep-learning model. This is not an artefact: the switching threshold is defined by instantaneous state (Speed, Gear), so models that focus on instantaneous state generalise better than those that accumulate history.

3.5 Circuit-by-Circuit Breakdown and Collapse Analysis

Model	Monaco	Monza	Silverstone	Suzuka	Mean
LR-instant	0.980	0.932	0.970	0.969	0.963
RF-instant	0.955	0.963	0.972	0.945	0.959
RF-lag	0.907	0.947	0.963	0.961	0.945
CausalTransformer	0.956	0.871	0.961	0.857	0.911
CausalLSTM	0.927	0.863	0.957	0.765	0.878
CausalGRU	0.728 ↓	0.896	0.968	0.917	0.877
TCN	0.944	0.890	0.968	0.537 ↓	0.835
CNN1D	0.811	0.821	0.956	0.684	0.818

Table 13. AUC-ROC per held-out circuit. ↓ marks circuit-specific collapses.

The GRU achieves AUC = 0.728 at Monaco — near random — yet 0.968 at Silverstone under identical training. LR-instant scores 0.980 on the same Monaco test set, so the failure is specific to the GRU's learning mechanism. The root cause is circuit speed memorisation: during training on the faster circuits, the GRU's hidden state learns that X-Mode is preceded by speeds building toward 280–360 km/h. Monaco never exceeds ~ 210 –220 km/h, so that signature never appears and the GRU predicts X-Mode for only 0.5% of Monaco windows (F1-X-Mode = 0.005), despite Monaco genuinely having 13% X-Mode frames. TCN exhibits the same failure mode at Suzuka.

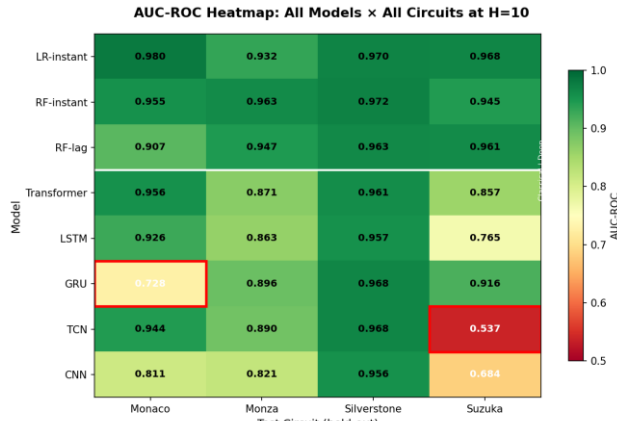


Figure 34. LOCO heatmap — GRU collapses at Monaco, TCN at Suzuka.



Figure 35. Monaco breakdown — GRU far below all other models.

Circuit memorisation failure. The GRU learned a speed-trajectory signature for X-Mode from fast circuits. At Monaco that signature never appears, so it effectively predicts Z-Mode for the entire race (F1-X-Mode = 0.005). The same mechanism explains the TCN collapse at Suzuka.

3.6 Prediction Horizon Sweep (H = 1 to 50)

All models were evaluated at horizons of 0.1, 0.5, 1.0, 2.5 and 5.0 seconds. Performance starts near 1.000 at H = 1 (the car barely changes in 0.1 s) and degrades with horizon, approaching chance at H = 50. H = 10 (one second) is the practical sweet spot: a meaningful gap from chance while remaining actionable for wing control.

Model	H=1	H=5	H=10	H=25	H=50
LR-instant	0.999	0.993	0.963	0.786	0.471
RF-instant	0.999	0.993	0.959	0.777	0.562
RF-lag	0.999	0.985	0.945	0.745	0.466
CausalTransformer	0.999	0.987	0.911	0.691	0.595
CausalGRU	0.999	0.986	0.877	0.682	0.505

Table 14. Mean AUC-ROC across circuits by prediction horizon.

Critically, the GRU's Monaco AUC is already low at H = 1 (0.753) — the collapse is circuit-specific, not horizon-specific, confirming the problem lies in the model's circuit representation rather than long-horizon difficulty.

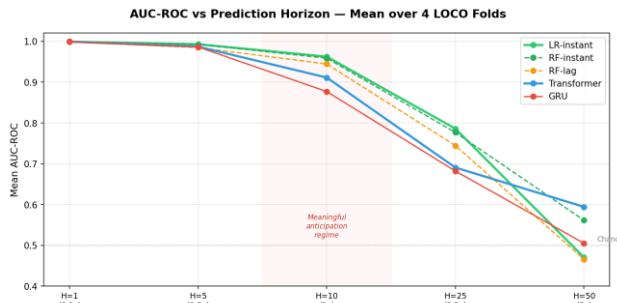


Figure 36. AUC-ROC vs horizon — H = 10 is the anticipation sweet spot.

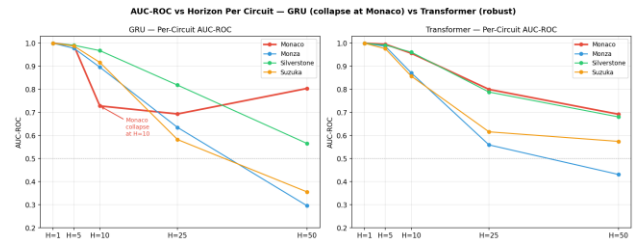
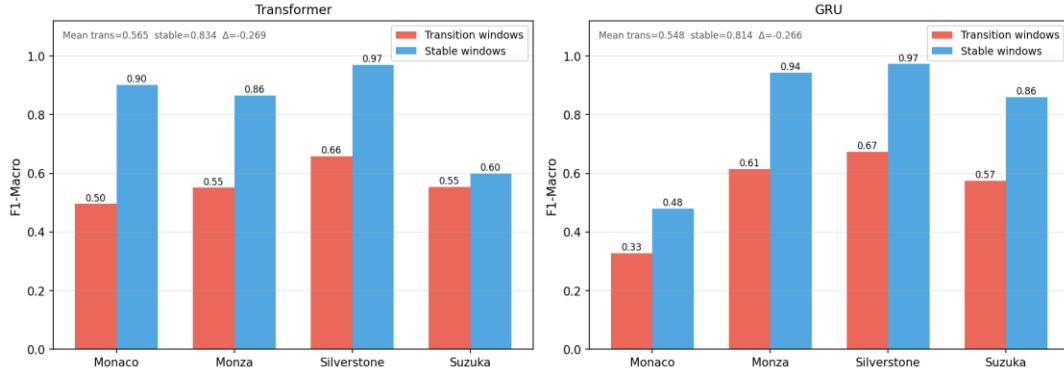


Figure 37. GRU (collapses at Monaco) vs robust Transformer across horizons.

3.7 Mode-Change Transition Window Analysis

A transition window is the ± 5 -frame (± 0.5 s) region around each mode switch, where prediction is hardest because the sensor state has not yet reflected the impending change. The F1 drop at transition windows is a remarkably consistent -0.27 across all circuits and both deep models, establishing it as an inherent task difficulty rather than a model-specific deficiency — and motivating transition-weighted loss as future work.

Circuit	Model	F1 Stable	F1 Transition	Drop
Monaco	Transformer	0.902	0.496	-0.406
Monza	Transformer	0.865	0.551	-0.314
Silverstone	Transformer	0.970	0.659	-0.311
Suzuka	Transformer	0.599	0.554	-0.045
Average	Transformer	0.834	0.565	-0.269
Average	GRU	0.814	0.548	-0.266

Table 15. Stable vs transition-window F1 — a consistent ≈ -0.27 drop.Transition-Window vs Stable-Window F1-Macro at H=10 ($\pm 0.5s$ around mode changes)Figure 38. F1-Macro at stable vs transition windows — a consistent ~ 0.27 gap across circuits.

3.8 Feature Interpretability — Integrated Gradients and Attention

To diagnose the divergence in cross-circuit generalisation, Integrated Gradients (Sundararajan et al., 2017) quantified each feature's attribution, and causal attention maps were extracted from the Transformer. At Monaco the two models rely on almost entirely different feature sets: the GRU weights dynamic, trajectory-dependent features (Longitudinal Force, Acceleration) calibrated to fast circuits, while the Transformer weights instantaneous-state features (Throttle, Speed, nGear) that remain valid regardless of the circuit's speed envelope — the same features LR-instant uses.

Rank	GRU (Monaco)	IG	Transformer (Monaco)	IG
1	Longitudinal Force	0.099	Throttle	0.120
2	Acceleration	0.087	Speed	0.108
3	Throttle	0.072	nGear	0.097
4	Speed	0.065	Kinetic Energy	0.084
5	Kinetic Energy	0.054	Longitudinal Force	0.063

Table 16. Integrated Gradients at Monaco — GRU uses dynamics, Transformer uses state.

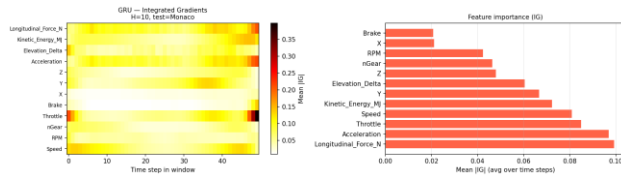


Figure 39. GRU IG at Monaco — trajectory features mislead.

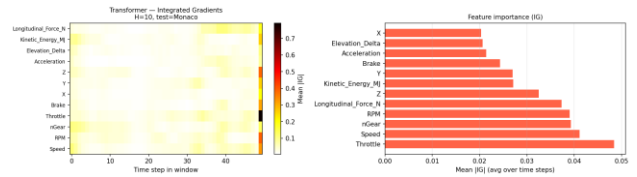


Figure 40. Transformer IG at Monaco — instantaneous state dominates.

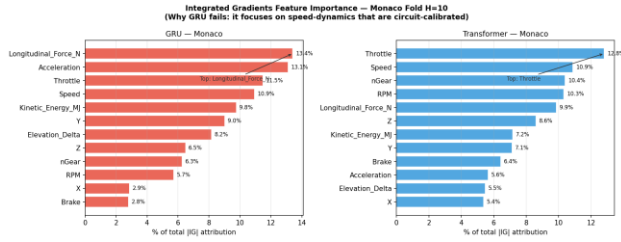


Figure 41. GRU vs Transformer feature importance side-by-side (Monaco).

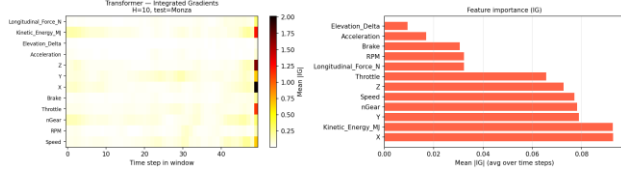


Figure 43. Transformer IG at Monza — consistent state features.

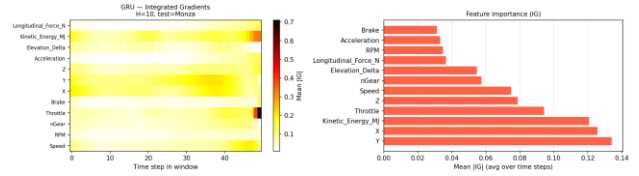


Figure 42. GRU IG at Monza — same features now correctly calibrated.

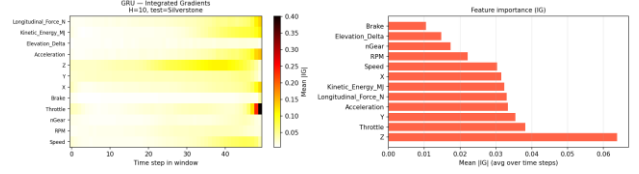


Figure 44. GRU IG at Silverstone — best GRU result; well-calibrated.

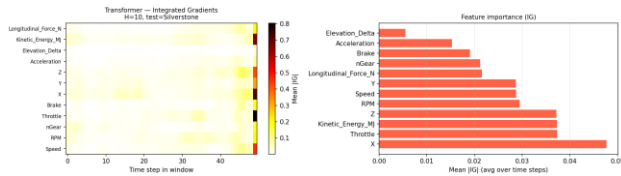


Figure 45. Transformer IG at Silverstone — instantaneous features lead.

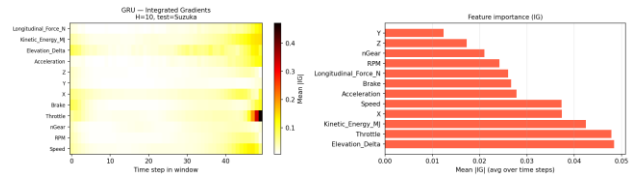


Figure 46. GRU IG at Suzuka — partial match to training speed profile.

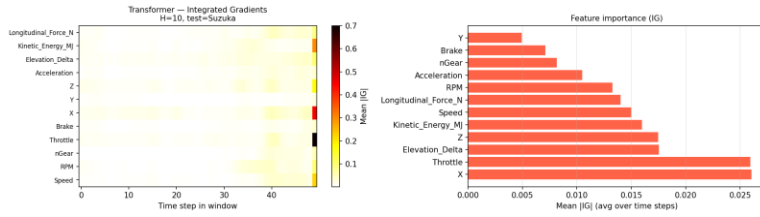


Figure 47. Transformer IG at Suzuka — Throttle and Speed remain the top features.

The Transformer's averaged causal attention maps attend broadly to recent frames without fixating on a speed-trajectory pattern, and this structure is consistent across all four circuits — confirming a circuit-agnostic temporal strategy that explains its robust generalisation.

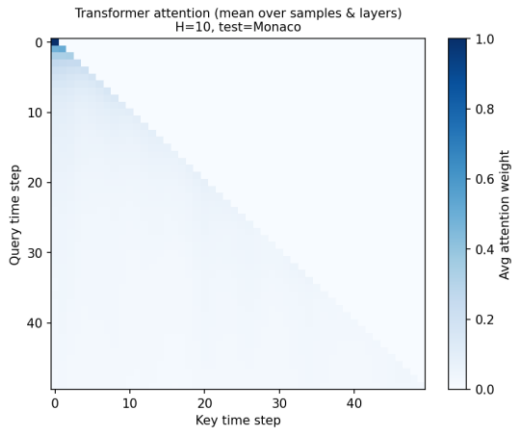


Figure 48. Causal attention — Monaco.

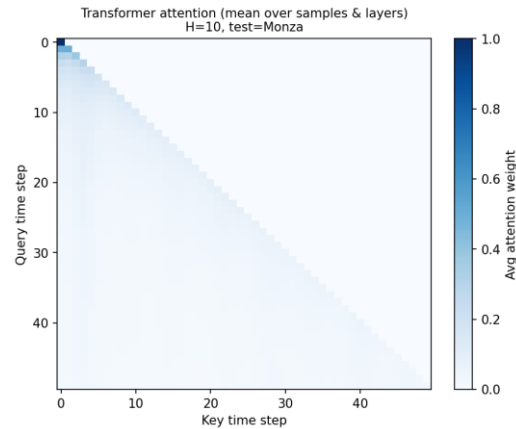


Figure 49. Causal attention — Monza.

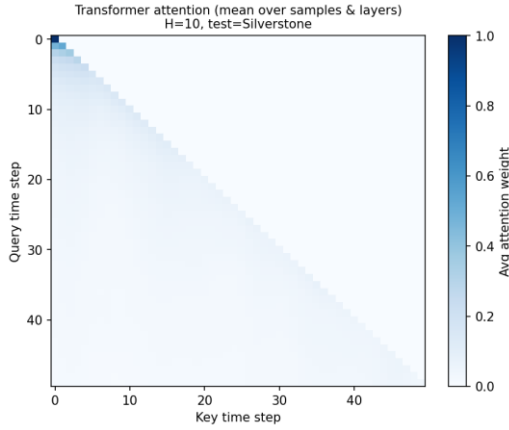


Figure 50. Causal attention — Silverstone.

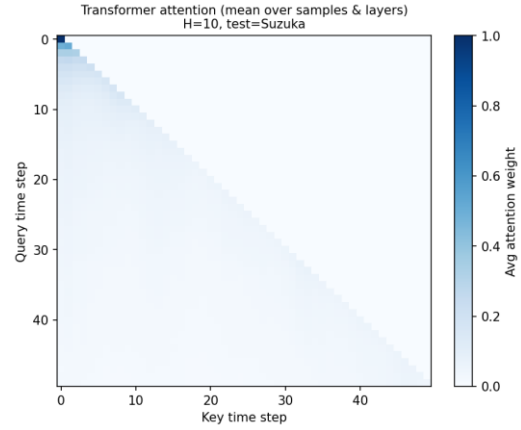


Figure 51. Causal attention — Suzuka.

3.9 Statistical Significance — Bootstrap Confidence Intervals

With only four LOCO test circuits, parametric tests are unreliable, so a non-parametric bootstrap ($n = 10,000$) constructed 95% CIs for the difference in mean AUC-ROC between LR-instant and each model; a CI excluding zero indicates significance. LR-instant is statistically superior to all deep models, while its advantage over RF-instant and RF-lag is not distinguishable — the three classical models are effectively equivalent at this evaluation scale.

Comparison	Mean Δ AUC	95% Bootstrap CI	Significant?
LR-instant vs RF-instant	+0.004	[-0.017, +0.024]	No
LR-instant vs RF-lag	+0.018	[-0.010, +0.057]	No
LR-instant vs CausalTransformer	+0.051	[+0.017, +0.090]	Yes
LR-instant vs CausalLSTM	+0.085	[+0.027, +0.166]	Yes
LR-instant vs CausalGRU	+0.086	[+0.015, +0.198]	Yes
LR-instant vs TCN	+0.128	[+0.012, +0.332]	Yes
LR-instant vs CNN1D	+0.145	[+0.053, +0.241]	Yes

Table 17. Bootstrap significance of LR-instant vs each model.

4. Synthesis — Connecting Phase 1 and Phase 2

The two phases form a coherent narrative rather than two independent experiments. Phase 1 established that the current mode is perfectly detectable from instantaneous features, and that Suzuka's anomalies are moments where the speed trajectory deviates sharply from normal (the maximum-braking events at tight corners). Phase 2 revealed that the GRU collapses at Monaco because Monaco's speed profile deviates sharply from the training circuits — structurally the same idea: the GRU treats Monaco as an "anomalous" environment because its speed envelope does not match the learned representation of an F1 race.

The models that survive cross-circuit generalisation (LR-instant, Transformer) share one property: they evaluate instantaneous state rather than accumulating a speed history, so they never form an expectation about what speed level is "normal." The practical implication for active-aero control design is direct: the control algorithm should monitor instantaneous state variables (current speed, gear, throttle, elevation gradient) rather than accumulated speed trajectories. The switching rule is fundamentally a state rule, not a trajectory rule.

5. Conclusion

This study investigated machine learning for predicting the required aerodynamic mode of 2026 Formula 1 cars one second in advance, aiming to eliminate the wing-transition latency inherent in reactive systems.

Phase 1 demonstrated that the current mode is detectable with perfect accuracy (Random Forest AUC = 1.000) from instantaneous telemetry. Isolation Forest located anomalous moments ($\approx 5\%$ of frames) at the circuit's maximum-braking zones, K-Means produced four statistically significant driving-state clusters ($\eta^2 = 0.555$), and permutation analysis identified gear position as the single most informative feature.

Phase 2 conducted a strict LOCO evaluation of eight models across four circuits (159,712 frames). The principal finding is that Logistic Regression achieves the highest mean AUC-ROC of 0.963 ± 0.021 , statistically outperforming all seven more complex models. The CausalTransformer is the strongest deep model (0.911 ± 0.055), while CausalGRU and TCN suffer circuit-specific collapses (GRU at Monaco 0.728; TCN at Suzuka 0.537) attributable to circuit speed memorisation. Integrated Gradients confirmed that the GRU relies on circuit-specific dynamic features while the Transformer and LR-instant rely on universally valid instantaneous-state features. Transition windows exhibit a consistent -0.27 F1 drop, an inherent task characteristic.

The central lesson is that cross-domain temporal prediction benefits from features that capture instantaneous physics rather than accumulated history, because accumulated history is calibrated to the training domain and fails to transfer when the test domain operates in a different regime.

6. Future Work

1. **Domain-relative feature normalisation.** Replace absolute speed with speed relative to the circuit's running average or the session's 240-km/h percentile rank, presenting the GRU with a circuit-invariant signal that may eliminate the Monaco collapse.
2. **Adversarial domain generalisation.** Add a circuit discriminator the feature extractor is trained to fool, penalising circuit-specific encoding and enforcing circuit-agnostic representations.
3. **Transition-weighted loss.** Up-weight frames within ± 5 of a mode switch; the consistent -0.27 F1 drop at transitions is the largest remaining gap and the most practically significant moment for control.
4. **Full-calendar evaluation.** Extend LOCO to all 24 circuits of the 2025/2026 calendar to stress-test generalisation and identify circuit archetypes where models fail systematically.
5. **Real-time hardware deployment.** Profile inference latency of the smallest models (LR-instant: 13 params; Transformer: 68k) on automotive-grade MCUs toward a production anticipatory controller.
6. **Driver-in-the-loop validation.** Simulator testing with professional drivers to assess whether one-second anticipation produces measurable lap-time or balance improvements.

7. References

1. FIA. 2026 Formula One Technical Regulations, Article 3.10: Active Aerodynamic Systems. FIA, 2025.
2. Oehrly, T. FastF1: A Python package for accessing F1 data. GitHub, 2024.
3. Liu, F.T., Ting, K.M., and Zhou, Z.H. (2008). Isolation Forest. Proc. IEEE ICDM, 413–422.
4. Lin, T.Y., Goyal, P., Girshick, R., He, K., and Dollár, P. (2017). Focal Loss for Dense Object Detection. Proc. IEEE ICCV, 2980–2988.
5. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. NeurIPS 30.
6. Bai, S., Kolter, J.Z., and Koltun, V. (2018). An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modelling. arXiv:1803.01271.
7. Sundararajan, M., Taly, A., and Yan, Q. (2017). Axiomatic Attribution for Deep Networks. Proc. ICML 34, 3319–3328.
8. Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
9. Cho, K., van Merriënboer, B., Gulcehre, C., et al. (2014). Learning Phrase Representations using RNN Encoder-Decoder for SMT. Proc. EMNLP.
10. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
11. Efron, B. and Tibshirani, R.J. (1993). An Introduction to the Bootstrap. Chapman & Hall/CRC.

12. Loshchilov, I. and Hutter, F. (2019). Decoupled Weight Decay Regularisation. Proc. ICLR 2019.
13. Scikit-learn developers. (2024). scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.
14. Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. NeurIPS 32.